Q1.

(a) The diagrams below show how much heat is lost from different parts of a house every second.



Through which part of the house above is most heat lost?

.....

1 mark

(b) Part of the house is insulated to reduce the loss of heat.
This is shown below.



(i) Which part of the house has been insulated?

.....

1 mark

(ii) Explain your answer.

.....

.....

1 mark

(c) The table below gives information about three fossil fuels that can be used to heat a house.

fuel	physical state	energy released when 1g is burned (J)	Does the fuel produce these substances when burned?	
			water	sulphur dioxide
coal	solid	25000	yes	yes
oil	liquid	42000	yes	yes
methane	gas	55000	yes	no

(i) Which fuel in the table releases the **least** energy when 1 g is burned?

.....

1 mark

(ii) Methane **can** be compressed.
Which information in the table shows that methane can be compressed?

.....

1 mark

(iii) Sulphur dioxide causes acid rain.
Use the table to explain why burning methane does **not** produce acid rain.

.....

.....

1 mark
maximum 6 marks

Q1.

(a) walls

1 (L3)

(b) (i) roof

1 (L3)

(ii) any **one** from

- it now loses 700 (J)
  accept 'it is only 700'
  'it is 700' is insufficient
- the energy is less (than before)
  accept 'it was 3 400 (J)'
- the energy or heat is different
  accept 'it has gone down'
- all the others do not change
  accept 'insulation reduces heat loss'
  'insulation keeps heat in' is insufficient

1 (L4)

(c) (i) coal

'solid' is insufficient
'25 000 J' is insufficient

1 (L4)

(ii) it is a gas

accept 'physical state'

1 (L4)

(iii) no sulphur dioxide (is given off)

accept 'it says no in the sulphur dioxide column'
*do **not** accept 'it has no sulphur dioxide in it'*
accept 'there is no sulphur in it'

1 (L4)

[6]